DESCRIPTION OF COURSEWORK

Course Code	G0251
Course Name	Foundations of Business Analytics
Lecturer	Sulfeeza Mohd Drus
Academic Session	2025/02
Assessment Title	Descriptive Statistics and Predictive Analytics

Title: Business Analytics and Data Mining Use Case

A. Introduction/ Situation/ Background Information

The goal of this assignment is to provide students with an opportunity to demonstrate their understanding of the given dataset by performing suitable descriptive analysis and identifying the relevant predictive analytics. Essentially, this assignment assesses both data exploration skills and the ability to link the dataset to predictive modeling approaches.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 2 Explain the analysis results of the identified business problem

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).

ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.

iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

This is a **GROUP** assignment, where each group must consist of **THREE (3) members** only. Each group needs to submit only **ONE (1) softcopy** submission in Moodle. The due date to submit this assignment is **TUESDAY, 11th MARCH 2025, 5pm**. Late submission (after 5pm) will **NOT BE ACCEPTED**.

Submission requirements:

1. Each group needs to submit a report (softcopy) by the deadline.
2. The report (with cover page and marking rubric) must be saved in PDF format.
3. Only group leaders need to submit the PDF file in Moodle.
4. Use the following file naming format (**GroupLeaderName_G0251 Assignment.pdf**).

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Knowledge	80
2.	Clarity of Writing and Organization	20
	TOTAL	**100**

F. Task(s)

1. You are given a dataset, **Bank Marketing.xlsx**. Please download the dataset from Moodle.
2. Based on your chosen dataset, complete the following tasks:
   a. **Perform Descriptive Analysis** – Summarize and explore the dataset using statistical measures (e.g., mean, median, mode, standard deviation) and visualizations (e.g., bar charts, line charts, histograms, scatter plots). Use appropriate exploratory techniques to understand the characteristics and patterns in the data.
   b. **Create a Data Dashboard** – Compile relevant visualizations into a single dashboard to provide a clear and concise overview of the data. The dashboard should help viewers interpret key insights effectively.
   c. **Identify Relevant Predictive Analytics** – Determine which predictive analytics techniques (e.g., classification, regression, time series forecasting) are appropriate for the dataset. Justify your choice by explaining how these models align with the dataset's features and potential use cases.

      * For Step 1, Step 2(a) and 2(b), you have the option to use either Excel or Tableau (Public).
3. Explain the following in your report:
   a. Descriptive analysis
      i. Present the results of statistical measures and explain how they describe the dataset.
      ii. Discuss the visualizations created—what insights they provide and why they are relevant to the dataset.

      * Please ensure that the charts are included in your report.

b. Dashboard
    i. Identify the main audience of the dashboard. Justify your answer.
    ii. List the charts included in the dashboard and justify their inclusion.
    iii. Explain how the audience can derive key insights from the dashboard.
c. Predictive Analytics
    i. Identify potential predictions that can be made using the dataset.
    ii. Suggest suitable predictive techniques (e.g., classification, regression, time series forecasting) for the identified predictions.
    iii. Explain how these predictions can benefit the company.

Format of the report:

i. *Cover Page*: Use the Assignment Cover Page Template
ii. *Section Title*: Font size – 12, Bold, 1.5 line spacing; Alignment – left-justified
iii. *Body*: Font size – 11, 1.5 line spacing; Alignment – justified
iv. *Table (optional)*: Font size – 10, single line spacing; Alignment – center
v. *Figure*: Alignment – center
vi. *Table Caption (only if your report has tables)*: Position – above the table; Font size – 10, single line spacing
vii. *Figure Caption*: Position – below the table; Font size – 10, single line spacing

APPENDIX 1

MARKING RUBRICS

Component	KNOWLEDGE (80%)					
Criteria	**Score and Descriptors**					**Marks**
	Excellent (9 - 10)	**Good (7 - 8)**	**Average (5 - 6)**	**Need Improvement (3 - 4)**	**Poor (1 - 2)**	
Descriptive Analysis	• Able to describe the results of statistical measurement clearly and concisely.	• Able to describe the results of statistical measurement adequately.	• Able to describe the results of statistical measurement in a basic manner.	• Limited ability to describe the results of statistical measurement.	• Unable to describe the results of statistical measurement.	
	• Able to describe the relevancy of the visualizations to the dataset clearly and concisely.	• Able to describe the relevancy of the visualizations to the dataset adequately.	• Able to describe the relevancy of the visualizations to the dataset in a basic manner.	• Limited ability to describe the relevancy of the visualizations to the dataset.	• Unable to describe the relevancy of the visualizations to the dataset.	
	• Able to describe the insights provided by the visualizations clearly and concisely.	• Able to describe the insights provided by the visualizations adequately.	• Able to describe the insights provided by the visualizations in a basic manner.	• Limited ability to describe the insights provided by the visualizations.	• Unable to describe the insights provided by the visualizations.	
Dashboard	• Able to identify and justify the main audience of the dashboard clearly and concisely.	• Able to identify and justify the main audience of the dashboard adequately.	• Able to identify and justify the main audience of the dashboard in a basic manner.	• Limited ability to identify and justify the main audience of the dashboard.	• Unable to identify and justify the main audience of the dashboard.	
	• Able to describe and justify the charts included in the dashboard clearly and concisely.	• Able to describe and justify the charts included in the dashboard adequately.	• Able to describe and justify the charts included in the dashboard in a basic manner.	• Limited ability to describe and justify the charts included in the dashboard.	• Unable to describe and justify the charts included in the dashboard.	
	• Able to justify how the main audience can derive key insights from the dashboard clearly and concisely.	• Able to justify how the main audience can derive key insights from the dashboard adequately.	• Able to justify how the main audience can derive key insights from the dashboard in a basic manner.	• Limited ability to justify how the main audience can derive key insights from the dashboard.	• Unable to justify how the main audience can derive key insights from the dashboard.	
Predictive Analytics	• Able to identify and describe suitable prediction and predictive analytics technique clearly and concisely.	• Able to identify and describe suitable prediction and predictive analytics technique adequately.	• Able to identify and describe suitable prediction and predictive analytics technique in a basic manner.	• Limited ability to identify and describe suitable prediction and predictive analytics technique.	• Unable to identify and describe suitable prediction and predictive analytics technique.	
	• Able to describe the benefits of the predictive analytics to the company clearly and concisely.	• Able to describe the benefits of the predictive analytics to the company adequately.	• Able to describe the benefits of the predictive analytics to the company in a basic manner.	• Limited ability to describe the benefits of the predictive analytics to the company.	• Unable to describe the benefits of the predictive analytics to the company.	

Component	CLARITY OF WRITING AND ORGANIZATION (20%)					
Criteria	**Score and Descriptors**					**Marks**
	Excellent (9 - 10)	**Good (7 - 8)**	**Average (5 - 6)**	**Need Improvement (3 - 4)**	**Poor (1 - 2)**	
Documentation format and presentation of work	• Well-written and highly organized content. • No grammar, spelling and sentence structure mistakes.	• Good writing with adequate content organization. • Occasional issues on grammar, spelling and sentence structure.	• Writing is clear but not properly organized. • Several issues on grammar, spelling and sentence structure.	• Averagely written with disorganized content. • Many grammars and spelling mistakes, some inappropriate sentence usage and formation.	• Poorly written. • Poor in grammar, countless spelling mistakes, many inappropriate sentence usage and formation.	
Demonstration of professionalism in the report	• Demonstrated a high level of professionalism in the submission and presentation of the report.	• Good effort has been put in the submission and presentation of the report.	• Acceptable demonstration of efforts in the submission and presentation of the report.	• Poor demonstration of effort with limited professionalism displayed in the report submitted.	• Extremely poor and no professionalism displayed in the report submitted. • Plagiarism is clearly evident.	